

RUHIN PATEL

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EDUCATION

Stony Brook University, New York, United States of America

August 2023 - May 2025

Master of Science in Data Science

Relevant Coursework: Data Analysis, Big Data Analysis, Big Data Algorithms & Networks, Data Science Fundamentals, Data Management

Pandit Deendayal Energy University, Gandhinagar, India

July 2018 - May 2022

Bachelor of Technology in Information and Communication Technology

Relevant Coursework: Analysis of Algorithms, Artificial Intelligence, Machine Learning, OS, Signals & Systems, ADE, COAP, Object Oriented Programming

TECHNICAL SKILLS

- **Programming Languages & Databases:** Python, R, SQL, C++, MATLAB, C | PostgreSQL, MySQL, MongoDB, DynamoDB, Neo4j
- **ML & AI:** PyTorch, TensorFlow, Scikit-learn, BERT, Llama 2, GPT-3/4, NLP, Neural Networks, Coreference Models, LLM Fine-Tuning
- **Data & Cloud Tools:** Spark, PySpark, Hadoop, Pandas, NumPy, SQLAlchemy, Tree-sitter | AWS (EC2, S3, Lambda, SageMaker), Docker, Kubernetes
- **Dev & Workflow:** Git, GitHub Actions, Jenkins, REST APIs, Jupyter, Agile
- **Visualization & BI:** Tableau, Power BI, Matplotlib, Seaborn, Excel

PROFESSIONAL EXPERIENCE

IACS, Stony Brook University

Stony Brook, New York

Senior Research Aide

February 2025 – Present

- Fine-tuned a 7B parameter CodeLlama model using Hugging Face Transformers and PyTorch to specialize code generation for MADNESS (C++ scientific simulation), enhancing model accuracy on complex multiresolution algorithms.
- Constructed and optimized a high-quality C++ dataset by automating function-level code extraction with Tree-sitter and Python, processing over 10,000 MADNESS files on the SeaWulf HPC cluster for efficient model training.
- Streamlined training workflows on GPU infrastructure, leveraging SeaWulf's distributed computing resources to achieve a 98% reduction in preprocessing time and significantly accelerate LLM fine-tuning for scientific code.

Yoloh

Columbus, Ohio (Remote)

Data Scientist Intern

May 2024 – January 2025

- Engineered a robust Optical Character Recognition solution for insurance management by integrating Llama 2, GPT-3, and GPT-4 models, achieving a 95% accuracy rate in data extraction and automating 80% of policy processing tasks.
- Designed and implemented a scalable database architecture using DynamoDB and Neo4j, streamlining user response storage and enabling personalized communication through AWS services, resulting in a 40% improvement in customer satisfaction.
- Implemented an AWS Lambda function to unify and deliver user responses in JSON format, cutting overall processing latency by 50% and boosting reporting speed for BI pipelines.

Blackcoffer

Delhi, India (Remote)

Data Scientist Intern

February 2022 – May 2023

- Created a Python-based solution integrated with the Twitter API to provide actionable insights, boosting the client's marketing and social media impact by 20%.
- Automated the conversion of raw data into structured Excel sheets, reducing manual input effort by 70% and accelerating transformation cycles by 35%.
- Delivered high-quality projects ahead of schedule by collaborating with a team of 5 developers, contributing to a 5% increase in company revenue.

ACADEMIC PROJECTS

Detecting and Mitigating Gender Bias in NLP Systems

- Fine-tuned BERT-based coreference resolution models, achieving a 21.1% reduction in gender bias (F1 score difference) on the WinoBias dataset, aligning with benchmarks established by Zhao et al. (2018) .
- Implemented debiasing techniques such as gender swapping and debiased word embeddings, maintaining a 95% performance score on standard benchmarks while effectively reducing societal stereotypes in NLP systems.
- Devised a comprehensive bias detection framework integrating augmented datasets and advanced evaluation metrics, enhancing the inclusivity and trustworthiness of coreference resolution applications by 30%, as validated by improved fairness metrics.

Analysis of Student Stress Factors and Their Impact on Self-Esteem

- Conducted a comprehensive analysis using multiple linear regression in R, identifying academic performance and study load as key predictors, with the model explaining 69.57% of the variance in stress levels, demonstrating a strong fit.
- Evaluated the impact of social support on self-esteem through ANOVA, revealing statistically significant positive effects, with post-hoc Tukey analysis providing detailed insights into group differences, improving understanding of social dynamics by 25%.
- Benchmarked findings against existing literature, noting a significant negative correlation between self-esteem and academic stress ($r = -0.522$), consistent with studies highlighting the inverse relationship between these variables .

Football Analytics Using Spark

- Designed a Spark-based analytics system processing over 500,000 events for performance insights; built Python-based predictive models achieving 75% accuracy—outperforming baseline sports prediction models (~63%).
- Produced interactive data visualizations with Matplotlib and Seaborn, delivering actionable insights into player performance, team strategies, and league-wide trends for data-driven decision-making.

CERTIFICATIONS

- **Professional Machine Learning Engineer:** Google
- **Google Cloud Professional Data Engineer Certification Course:** Udemy
- **Data Structures A-Z: Data Structures + Algorithms Bootcamp:** Udemy
- **IBM Data Science Professional Certificate:** Coursera